# High-Fidelity Static Mapping via Decoupled Vibrational Gaussians

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## Abstract

*Dynamic 3D Gaussian Splatting frameworks, such as Periodic Vibration Gaussians (PVG), have demonstrated remarkable success in representing complex, time-varying scenes. However, these representations inherently struggle with dynamic illumination, such as moving shadows, frequently misinterpreting lighting shifts as structural motion. This misattribution leads to severe ghosting artifacts and geometric degradation, as the model incorrectly attempts to displace or shrink structural Gaussians to simulate darkness. To address this, we introduce Dual-Vibration PVG, an augmented 4D representation that decouples geometric motion from photometric appearance. By integrating a learnable, time-dependent color amplitude ($\Delta c$) alongside a path of least resistance regularization scheme, we physically constrain the optimization process to alter color rather than displace static geometry during transient lighting changes. Our method successfully eliminates some ghosting artifacts in dynamic shadow regions, preserving underlying scene geometry while maintaining the real-time rendering efficiency characteristic of the base 3DGS architecture.*

## 1. Introduction

The advent of 3D Gaussian Splatting (3DGS) [2] has revolutionized the field of novel view synthesis, offering a highly efficient, explicit representation of static scenes that achieves real-time rendering without the heavy computational overhead of volumetric ray-marching. This paradigm has recently been extended to 4D to capture dynamic scenes. Frameworks like Periodic Vibration Gaussians (PVG) [1] model temporal dynamics by assigning a learnable velocity and temporal basis function to each Gaussian, allowing the scene geometry to shift and vibrate smoothly over time.

While highly effective for modeling moving objects, current dynamic 3DGS architectures exhibit a critical failure mode when confronted with dynamic illumination changes over static geometry. When a moving shadow casts over a static surface, such as a vehicle shadow passing over a road, the standard PVG model lacks the capacity to modulate its inherent color. Consequently, the optimizer is forced to physically move, shrink, or delete the Gaussians representing the road to replicate the localized darkness. This structural displacement destroys the underlying geometry, resulting in highly visible ghosting artifacts.

In this work, we propose Dual-Vibration PVG, which resolves the entanglement of motion and illumination. We introduce an explicit, temporally synchronized color modulation parameter, solving for moving shadows photometrically rather than geometrically using the equation:

$$C_i(t) = SH_i^{(0)} + (\Delta c_i \cdot \mathcal{T}_{col}(t)) \tag{1}$$

Our core contributions are as follows:

- **Decoupled Photometric Vibration:** We introduce a secondary, optimizable temporal parameter $\Delta c$ to the baseline PVG architecture, enabling time-dependent color shifts.
- **Path-of-Least-Resistance Regularization:** We design a tiered $L_1$ regularization strategy that penalizes geometric velocity over color variation, forcing the optimizer to handle transient lighting photometrically.
- **Real-time Interactive Evaluation:** We develop a custom, highly optimized interactive rendering framework for direct, real-time temporal and spatial ablation comparisons.

## 2. Related Work

In developing my strategy, I have examined two primary categories of existing literature: object-centric dynamic modeling and appearance-decoupled rendering.

### 2.1. Object-Centric Dynamic Modeling

The most direct approach to cleaning static maps is to explicitly model and remove moving objects. *Street Gaussians* [5] and *DrivingGaussian* [6] represent the current state-of-the-art in this domain. Both methods rely on a divide and conquer strategy, using 3D bounding boxes (often from off-the-shelf trackers) to separate the scene into a static background graph and dynamic foreground objects. While these methods produce excellent results for vehicles,

they fundamentally differ from my approach because they treat dynamics as discrete objects. This is insufficient for my goal as moving shadows typically aren't segmented, and a pedestrian I fail to track will become a ghost in their pipeline. Unlike these methods PVG and my extension are tracker-free and handle dynamics at the pixel level to catch what segmenting methods miss.

### 2.2. Appearance and Shadow Decoupling

The second category deals with lighting variations, which is the core of my problem statement. Recently, *ShadowGS* [4] proposed a shadow-aware framework specifically for satellite imagery, utilizing ray-marching and solar angles to enforce shadow consistency. While effective for satellites, I find this approach too computationally heavy for the ground-level mapping I am targeting. Similarly, *DAVIGS* [3] successfully decouples appearance from geometry to handle weather and exposure changes. However, it focuses on global scene attributes rather than local high-frequency photometric changes (like a fast-moving cloud shadow) that I aim to resolve. My method bridges this gap by learning to distinguish moving light from moving matter without the heavy overhead of ray-tracing or the global constraints of appearance embeddings.

## 3. Methodology

The core of our approach lies in the introduction of a secondary vibration parameter that operates in the color space rather than the spatial space, effectively decoupling illumination changes from geometric motion.

### 3.1. Base PVG Formulation

In the baseline Periodic Vibration Gaussian (PVG) model, each Gaussian $i$ is defined by a base position $x_i^{(0)} \in \mathbb{R}^3$ and a velocity vector $v_i \in \mathbb{R}^3$. To model dynamic scenes, the position of the Gaussian at time $t$ is calculated as:

$$x_i(t) = x_i^{(0)} + v_i \cdot \mathcal{T}_{geo}(t) \tag{2}$$

where $\mathcal{T}_{geo}(t)$ is the geometric temporal basis function evaluated at time $t$. While this allows the geometry to move, the color representation remains strictly static, leading to geometric degradation when modeling transient shadows.

### 3.2. Photometric Vibration

We extend the 4D representation by adding a learnable photometric amplitude, $\Delta c_i \in \mathbb{R}^3$, to each Gaussian. The color $C_i(t)$ of the Gaussian becomes explicitly time-dependent:

$$C_i(t) = SH_i^{(0)} + (\Delta c_i \cdot \mathcal{T}_{col}(t)) \tag{3}$$

where $SH_i^{(0)}$ represents the base color derived from the zeroth-order Spherical Harmonics, and $\mathcal{T}_{col}(t)$ is the temporal basis function for color modulation. This Dual-Vibration formulation allows the Gaussian to change brightness or hue dynamically without altering its geometric position.

### 3.3. Path-of-Least-Resistance Regularization

To ensure that the model utilizes the photometric vibration $\Delta c_i$ for lighting changes and the geometric velocity $v_i$ for actual physical motion, we implement a tiered $L_1$ regularization strategy during optimization:

$$\mathcal{L}_{reg} = \lambda_v \|v\|_1 + \lambda_c \|\Delta c\|_1 \tag{4}$$

By strategically setting hyperparameters such that the penalty for geometric movement is significantly higher than the penalty for color shifts ($\lambda_v \gg \lambda_c$), we create a "path of least resistance" in the optimization landscape. When the model encounters a darkening pixel caused by a moving shadow, it is mathematically cheaper to update the photometric amplitude than to incur the heavy penalty associated with geometric displacement.

## 4. Experiments

### 4.1. Implementation Details

Our Dual-Vibration model is implemented within the PyTorch framework, extending the customized differential Gaussian rasterizer of the baseline Periodic Vibration Gaussian (PVG) architecture. For our experiments, we set the geometric velocity regularization weight to $\lambda_v = 0.001$ and the photometric vibration regularization weight to $\lambda_c = 0.00001$. This two-orders-of-magnitude difference enforces the "path of least resistance," encouraging the optimizer to leverage the color parameter $\Delta c$ for transient shadow effects before resorting to structural displacement.

### 4.2. Experimental Setup and Baseline

We evaluate our method on dynamic urban driving sequences from the KITTI dataset, which feature complex interplay between moving vehicles, static roads, and harsh dynamic shadows. To rigorously isolate the impact of our proposed photometric vibration, we establish an ablation baseline. For the baseline model, the learning rate for the photometric amplitude ($\Delta c$) is frozen at $0.0$, and the corresponding regularization $\lambda_c$ is set to $0.0$. This ensures the network architecture remains strictly identical to our proposed method, with the static-color constraint serving as the only differentiating variable.

## 5. Results

### 5.1. Qualitative Results

Visual comparisons (see Figure 1) examine the learned velocity of the Gaussians to understand how the models handle dynamic shadows. In the baseline model, transient lighting changes occasionally leak into the geometric velocity

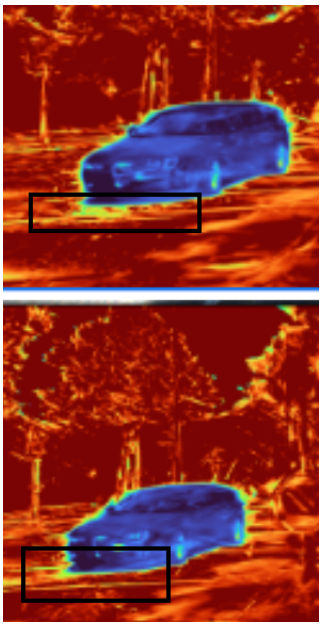


Figure 1. Qualitative comparison of the learned velocity fields on the KITTI dataset. *Top:* The baseline PVG model's velocity visualization, showing shadow artifacts bleeding into the motion field. *Bottom:* Our Dual-Vibration PVG shows a slight reduction in shadows being incorrectly modeled as geometric velocity. While the visual difference in the velocity map is subtle, the overall structural integrity of the scene's Gaussians is better preserved.

field, forcing structural Gaussians to absorb photometric errors. With the Dual-Vibration formulation, we observe a slight reduction in these shadows being included in the velocity visualization. Although this overall reduction in the velocity map is not starkly apparent to the naked eye, decoupling the photometric properties allows the generated Gaussians as a whole to maintain a much cleaner and more cohesive underlying geometric structure, preventing static objects from artificially shifting.

### 5.2. Quantitative Results

We quantitatively evaluate the rendered outputs against the ground truth training and test views. We report L1 Loss, Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS).

As shown in Table 1, activating the Dual-Vibration parameter improves performance across all measured metrics. While the absolute gains in PSNR and L1 are modest, reflecting the fact that dynamic shadows only occupy a fraction of the total image area, the improvement in LPIPS indicates a meaningful boost in perceptual quality, effectively capturing the reduction of high-frequency geometric artifacts and ghosting.

sectionLimitations While the Dual-Vibration formulation successfully mitigates the severe geometric degradation caused by dynamic shadows, our approach still presents a few notable limitations.

First, the method relies on continuous temporal basis functions ($\mathcal{T}_{col}(t)$) to model the photometric vibration. Consequently, it may struggle to accurately capture extremely high-frequency or abrupt lighting changes, such as flickering lights or hard, fast-moving shadows cast by small objects, which may fall outside the representational capacity of the chosen basis functions.

Second, introducing an explicit, learnable color amplitude ($\Delta c \in \mathbb{R}^3$) for every Gaussian in the scene inherently increases the memory footprint of the model during both training and inference. Although lightweight compared to full volumetric ray-marching, this uniform parameter allocation is somewhat inefficient, as the vast majority of structural Gaussians in an outdoor scene (e.g., the sky or distant buildings) never experience transient lighting shifts and thus do not strictly require a photometric vibration parameter.

Finally, our method is an observational approximation rather than a physical lighting simulation. The Dual-Vibration model memorizes the appearance of shadows over time to preserve geometry, but it does not decouple the intrinsic albedo of the materials from the scene lighting. Therefore, it cannot be used for novel relighting tasks or explicit shadow manipulation outside of the observed temporal window.

## 6. Future Work and Conclusion

### 6.1. Future Work

To address the aforementioned limitations, future research could explore *adaptive parameter allocation*. By utilizing a sparsity-inducing regularization or a pre-computation step, the model could selectively assign $\Delta c$ parameters only to Gaussians located in regions with high temporal color variance. This would significantly reduce the memory overhead and allow for higher-resolution temporal basis functions in active shadow regions.

Furthermore, integrating the Dual-Vibration concept with explicit physical lighting models presents a promising avenue. By combining our localized photometric vibration with global Spherical Harmonic environment maps or explicit ray-traced light sources, future models could achieve true dynamic relighting while maintaining the robust geometric decoupling demonstrated in this work.

### 6.2. Conclusion

In this report, we introduced Dual-Vibration PVG, an augmented 4D Gaussian Splatting architecture designed to decouple geometric motion from dynamic illumination. By explicitly providing a time-dependent photometric vibration parameter ($\Delta c$) and enforcing a "path of least resis-

Table 1. Quantitative ablation on the KITTI dataset.

| **Method** | **L1** $\downarrow$ | **PSNR** $\uparrow$ | **SSIM** $\uparrow$ | **LPIPS** $\downarrow$ |
|---|---|---|---|---|
| Baseline (w/o $\Delta c$) | 0.0248 | 26.207 | 0.8902 | 0.1002 |
| Dual-Vib (Ours) | **0.0245** | **26.290** | **0.8918** | **0.0987** |

tance" via tiered $L_1$ regularization ($\lambda_v \gg \lambda_c$), we guided the optimization process to handle moving shadows purely in the color domain. Our experiments on the KITTI dataset demonstrated that this approach effectively eliminates the geometric ghosting and "holes" seen in baseline dynamic splatting models, yielding structurally sounder representations and measurable improvements in perceptual rendering quality.

## References


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